

Human Obstacle Detector and Tracker for Acme Robotics

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1. OVERVIEW & OBJECTIVES	2. APPROACH & METHODOLOGY
Project: Human Obstacle Detector and Tracker Duration: 3 weeks (Oct-Nov 2024) - Real-time human detection using YOLOv8 - Persistent tracking with unique IDs 3D position estimation in robot frame 90%+ code coverage with TDD	Architecture: - Preprocessor: Image normalization for DNNs - DetectorTracker: YOLO + IoU-based tracking - MLDepthEstimator: Monocular depth prediction - Transformer3D: 2D→3D coordinate transformation Process: - Agile Iterative Process (AIP) - Test-Driven Development (TDD) - Pair Programming with role rotation - CI/CD with GitHub Actions & CodeCov
3. TECHNICAL DETAILS	4. SCHEDULE & DELIVERABLES
Technologies: - C++17, CMake, OpenCV 4.x - YOLOv8 ONNX models (COCO dataset) - MiDaS/Depth Anything depth networks - Google Test, cppcheck, Doxygen Key Algorithms: - YOLO object detection - IoU-based multi-object tracking - Monocular depth estimation - Pinhole camera projection model	Schedule: - Phase 0 (Week 1): Design, UML, documentation - Phase 1 (Week 2): Core implementation (Sprints 1-2) - Phase 2 (Week 3): Integration & testing (Sprint 3) Deliverables: ✓ UML diagrams & documentation ✓ CI/CD pipeline with 90%+ coverage ✓ Production-ready C++ module ✓ Developer documentation ✓ Demo video